

10	(a)	According to wave theory, emission of electrons from the surface of the metal <u>can occur at any frequency</u> B[1] If exposure time is sufficiently long since energy can be accumulated B[1] For particulate model, electromagnetic radiation comes in discrete quantum/packet, where energy of each quantum/packet/photon proportional to frequency of radiation. B[1] Emission of electrons occur only if photon energy is greater than work function B[1]	
	(b)	(i)	$E_k = e\Delta V = 1.60 \times 10^{-19} \times 4700 = 7.52 \times 10^{-16} \text{ J} \quad \text{C1}$ $E_k = \frac{1}{2}mv^2 = \frac{p^2}{2m} \quad \text{C1}$ $\lambda = \frac{h}{\sqrt{2mE_k}} = \frac{6.63 \times 10^{-34}}{\sqrt{2 \times 9.11 \times 10^{-31} \times E_k}} \quad \text{M1}$ $\lambda = 1.8 \times 10^{-11} \text{ m} \quad \text{A1}$
		(ii)	De Broglie wavelength of electrons is of the same order of magnitude as the spacing between atoms in a crystal B1 Electron diffraction pattern reveals the arrangement of atoms within the crystal B1
	(c)	(i)	A photon is emitted when an electron transits from a higher energy level to a lower one. B1 Discrete lines showed that only photons of specific energies are emitted B1 Specific energy <u>change</u> implies discrete energy levels B1
		(ii)	(1) Arrow from -1.51 eV to -3.41 eV B1
		(2)	$\lambda = \frac{hc}{(3.41 - 0.38)e} \quad \text{C1}$ $\lambda = 4.10 \times 10^{-7} \text{ m} \quad \text{A1}$
		(iii)	continuous spectrum crossed by dark lines B1 four dark lines B1 corresponding to wavelengths of photons absorbed by hydrogen atom B1 absorbed photons re-emitted in all directions B1